

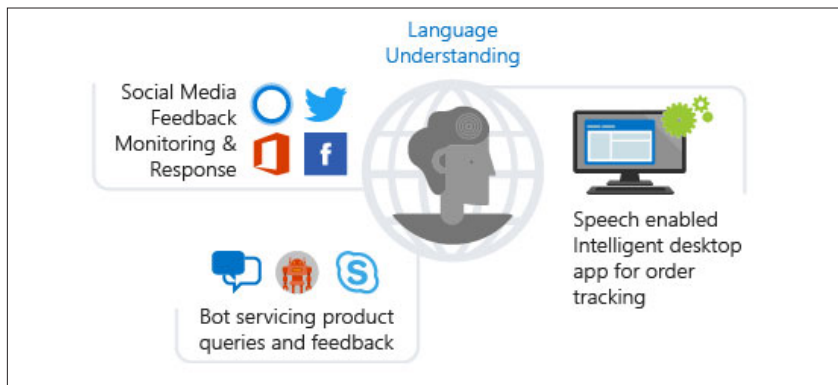


Figure 2: Azure LUIS and bot services (Image Source: Azure Documentation)

Azure Bot Service

Azure Language Understanding (LUIS) is a cloud API service from Microsoft, which uses custom ML services for conversational AI solutions like chatbot development. It uses an NLP technique called natural language understanding (NLU).

LUIS has three components – utterances, intent and entities. Utterances are plain text sentences which are thrown by users as a question to the bot. For example, “What time is the movie being shown in Casino?” Intents are the specific actions represented in the utterances. For example, ‘show time’ is the intent in the above example. Entity is the reference used in the utterance. ‘Casino theatre’ is the entity in this example.

LUIS can also be used for filtering negative or positive intents from social media posts.

When there is a user input as a sentence, LUIS takes the utterances and does NLU processing. It gets the intent score as a value between 0 and 1. Based on the best score fitment, it gets the entities associated and fetches the response for the same from the knowledge base.

LUIS can be used to create custom language processing capability for any local language by training the model to process new utterances of a custom language model. Also, there are built-in security features available to keep the LUIS API accessible in a secured way. LUIS can run on Azure cloud, on-premises or on the edge, as well as by

installing LUIS in a Dockerized container.

Amazon Lex

Amazon Lex is a service for building conversational interfaces into any application, using voice and text. It provides the advanced deep learning functionalities of automatic speech recognition (ASR) for converting speech to text, and natural language understanding (NLU) to recognise the intent of the text, to help build applications with highly engaging user experiences and lifelike conversational interactions

Amazon Lex enables any developer to build conversational chatbots quickly. No deep learning expertise is necessary—to create a bot, you just specify the basic conversation flow in the Amazon Lex console. It manages the dialogue and dynamically adjusts the responses in the conversation. Using the console, you can build, test, and publish your text or voice chatbot. You can then add the conversational interfaces to bots on mobile devices, web applications, and chat platforms (for example, Facebook Messenger).

Amazon Lex provides pre-built integration with AWS Lambda, and you can easily integrate with many other services on the AWS platform, including Amazon Cognito, AWS Mobile Hub, Amazon CloudWatch, and Amazon DynamoDB. Integration with Lambda

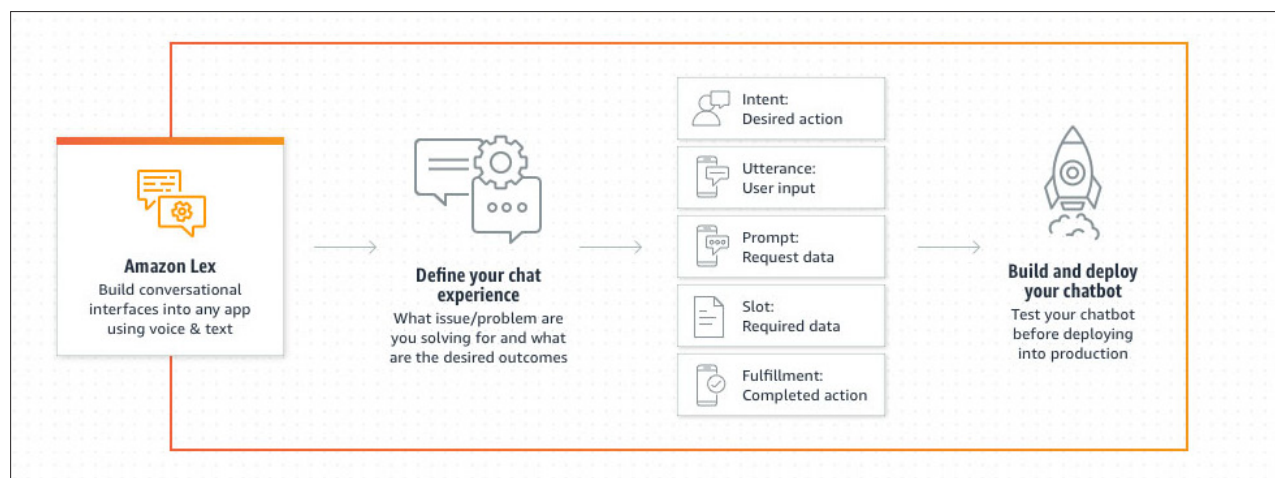


Figure 3: Amazon Lex